2
Risk-Free Assets

2.1 Time Value of Money

It is a fact of life that $100 to be received after one year is worth less than the same amount today. The main reason is that money due in the future or locked in a fixed term account cannot be spent right away. One would therefore expect to be compensated for postponed consumption. In addition, prices may rise in the meantime and the amount will not have the same purchasing power as it would have at present. Finally, there is always a risk, even if a negligible one, that the money will never be received. Whenever a future payment is uncertain to some degree, its value today will be reduced to compensate for the risk. (However, in the present chapter we shall consider situations free from such risk.) As generic examples of risk-free assets we shall consider a bank deposit or a bond.

The way in which money changes its value in time is a complex issue of fundamental importance in finance. We shall be concerned mainly with two questions:

> What is the future value of an amount invested or borrowed today?
>
> What is the present value of an amount to be paid or received at a certain time in the future?

The answers depend on various factors, which will be discussed in the present chapter. This topic is often referred to as the *time value of money*.

2.1.1 Simple Interest

Suppose that an amount is paid into a bank account, where it is to earn *interest*. The *future value* of this investment consists of the initial deposit, called the *principal* and denoted by P, plus all the interest earned since the money was deposited in the account.

To begin with, we shall consider the case when interest is attracted only by the principal, which remains unchanged during the period of investment. For example, the interest earned may be paid out in cash, credited to another account attracting no interest, or credited to the original account after some longer period.

After one year the interest earned will be rP, where $r > 0$ is the *interest rate*. The value of the investment will thus become $V(1) = P + rP = (1 + r)P$. After two years the investment will grow to $V(2) = (1 + 2r)P$. Consider a fraction of a year. Interest is typically calculated on a daily basis: the interest earned in one day will be $\frac{1}{365}rP$. After n days the interest will be $\frac{n}{365}rP$ and the total value of the investment will become $V(\frac{n}{365}) = (1 + \frac{n}{365}r)P$. This motivates the following rule of *simple interest*: The value of the investment at time t, denoted by $V(t)$, is given by

$$V(t) = (1 + tr)P, \tag{2.1}$$

where time t, expressed in years, can be an arbitrary non-negative real number; see Figure 2.1. In particular, we have the obvious equality $V(0) = P$. The number $1 + rt$ is called the *growth factor*. Here we assume that the interest rate r is constant. If the principal P is invested at time s, rather than at time 0, then the value at time $t \geq s$ will be

$$V(t) = (1 + (t - s)r)P. \tag{2.2}$$

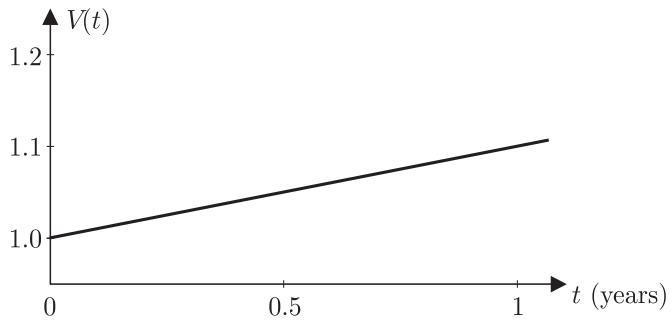


Figure 2.1 Principal attracting simple interest at 10% ($r = 0.1$, $P = 1$)

Throughout this book the unit of time will be one year. We shall transform any period expressed in other units (days, weeks, months) into a fraction of a year.

Example 2.1

Consider a deposit of \$150 held for 20 days and attracting simple interest at a rate of 8%. This gives $t = \frac{20}{365}$ and $r = 0.08$. After 20 days the deposit will grow to $V(\frac{20}{365}) = (1 + \frac{20}{365} \times 0.08) \times 150 \cong 150.66$.

The *return* on an investment commencing at time s and terminating at time t will be denoted by $K(s,t)$. It is given by

$$K(s,t) = \frac{V(t) - V(s)}{V(s)}. \tag{2.3}$$

In the case of simple interest

$$K(s,t) = (t-s)r,$$

which clearly follows from (2.2). In particular, the interest rate is equal to the return over one year,

$$K(0,1) = r.$$

As a general rule, interest rates will always refer to a period of one year, facilitating the comparison between different investments, independently of their actual duration. By contrast, the return reflects both the interest rate *and* the length of time the investment is held.

Exercise 2.1

A sum of $\$9,000$ paid into a bank account for two months (61 days) to attract simple interest will produce $\$9,020$ at the and of the term. Find the interest rate r and the return on this investment.

Exercise 2.2

How much would you pay today to receive $\$1,000$ at a certain future date if you require a return of 2%?

Exercise 2.3

How long will it take for a sum of \$800 attracting simple interest to become \$830 if the rate is 9%? Compute the return on this investment.

Exercise 2.4

Find the principal to be deposited initially in an account attracting simple interest at a rate of 8% if $\$1,000$ is needed after three months (91 days).

The last exercise is concerned with an important general problem: Find the initial sum whose value at time t is given. In the case of simple interest the answer is easily found by solving (2.1) for the principal, obtaining

$$V(0) = V(t)(1 + rt)^{-1}. \tag{2.4}$$

This number is called the *present* or *discounted value* of $V(t)$ and $(1 + rt)^{-1}$ is the *discount factor*.

Example 2.2

A *perpetuity* is a sequence of payments of a fixed amount to be made at equal time intervals and continuing indefinitely into the future. For example, suppose that payments of an amount C are to be made once a year, the first payment due a year hence. This can be achieved by depositing

$$P = \frac{C}{r}$$

in a bank account to earn simple interest at a constant rate r. Such a deposit will indeed produce a sequence of interest payments amounting to $C = rP$ payable every year.

In practice simple interest is used only for short-term investments and for certain types of loans and deposits. It is not a realistic description of the value of money in the longer term. In the majority of cases the interest already earned can be reinvested to attract even more interest, producing a higher return than that implied by (2.1). This will be analysed in detail in what follows.

2.1.2 Periodic Compounding

Once again, suppose that an amount P is deposited in a bank account, attracting interest at a constant rate $r > 0$. However, in contrast to the case of simple interest, we assume that the interest earned will now be added to the principal periodically, for example, annually, semi-annually, quarterly, monthly, or perhaps even on a daily basis. Subsequently, interest will be attracted not

just by the original deposit, but also by all the interest earned so far. In these circumstances we shall talk of *discrete* or *periodic compounding*.

Example 2.3

In the case of monthly compounding the first interest payment of $\frac{r}{12}P$ will be due after one month, increasing the principal to $(1+\frac{r}{12})P$, all of which will attract interest in the future. The next interest payment, due after two months, will thus be $\frac{r}{12}(1+\frac{r}{12})P$, and the capital will become $(1+\frac{r}{12})^2P$. After one year it will become $(1+\frac{r}{12})^{12}P$, after n months it will be $(1+\frac{r}{12})^nP$, and after t years $(1+\frac{r}{12})^{12t}P$. The last formula admits t equal to a whole number of months, that is, a multiple of $\frac{1}{12}$.

In general, if m interest payments are made per annum, the time between two consecutive payments measured in years will be $\frac{1}{m}$, the first interest payment being due at time $\frac{1}{m}$. Each interest payment will increase the principal by a factor of $1+\frac{r}{m}$. Given that the interest rate r remains unchanged, after t years the *future value* of an initial principal P will become

$$V(t) = \left(1+\frac{r}{m}\right)^{tm} P, \tag{2.5}$$

because there will be tm interest payments during this period. In this formula t must be a whole multiple of the period $\frac{1}{m}$. The number $\left(1+\frac{r}{m}\right)^{tm}$ is the *growth factor*.

The exact value of the investment may sometimes need to be known at time instants between interest payments. In particular, this may be so if the account is closed on a day when no interest payment is due. For example, what is the value after 10 days of a deposit of \$100 subject to monthly compounding at 12%? One possible answer is \$100, since the first interest payment would be due only after one whole month. This suggests that (2.5) should be extended to arbitrary values of t by means of a step function with steps of duration $\frac{1}{m}$, as shown in Figure 2.2. Later on, in Remark 2.6 we shall see that the extension consistent with the No-Arbitrage Principle should use the right-hand side of (2.5) for all $t \geq 0$.

Exercise 2.5

How long will it take to double a capital attracting interest at 6% compounded daily?

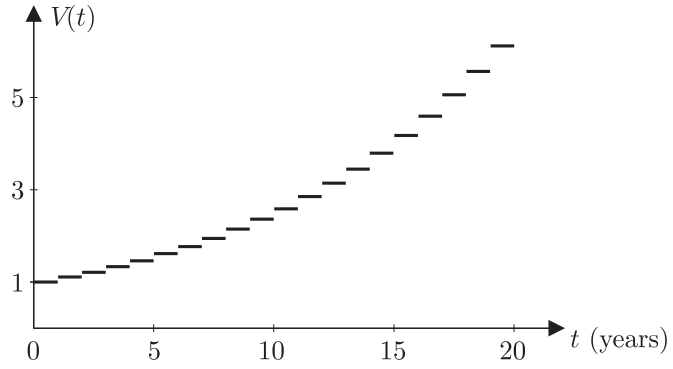


Figure 2.2 Annual compounding at 10% ($m = 1$, $r = 0.1$, $P = 1$)

Exercise 2.6

What is the interest rate if a deposit subject to annual compounding is doubled after 10 years?

Exercise 2.7

Find and compare the future value after two years of a deposit of \$100 attracting interest at a rate of 10% compounded a) annually and b) semi-annually.

Proposition 2.1

The future value $V(t)$ increases if any one of the parameters m, t, r or P increases, the others remaining unchanged.

Proof

It is immediately obvious from (2.5) that $V(t)$ increases if t, r or P increases. To show that $V(t)$ increases as the compounding frequency m increases, we need to verify that if $m < k$, then

$$\left(1 + \frac{r}{m}\right)^{tm} < \left(1 + \frac{r}{k}\right)^{tk}.$$

The latter clearly reduces to

$$\left(1 + \frac{r}{m}\right)^{m} < \left(1 + \frac{r}{k}\right)^{k},$$

which can be verified directly using the binomial formula:

$$\begin{aligned}\left(1+\frac{r}{m}\right)^m &= 1+r+\frac{1-\frac{1}{m}}{2!}r^2+\cdots+\frac{\left(1-\frac{1}{m}\right)\times\cdots\times\left(1-\frac{m-1}{m}\right)}{m!}r^m\\ &\leq 1+r+\frac{1-\frac{1}{k}}{2!}r^2+\cdots+\frac{\left(1-\frac{1}{k}\right)\times\cdots\times\left(1-\frac{m-1}{k}\right)}{m!}r^m\\ &< 1+r+\frac{1-\frac{1}{k}}{2!}r^2+\cdots+\frac{\left(1-\frac{1}{k}\right)\times\cdots\times\left(1-\frac{k-1}{k}\right)}{k!}r^k\\ &= \left(1+\frac{r}{k}\right)^k.\end{aligned}$$

The first inequality holds because each term of the sum on the left-hand side is no greater than the corresponding term on the right-hand side. The second inequality is true because the sum on the right-hand side contains $m-k$ additional non-zero terms as compared to the sum on the left-hand side. This completes the proof. □

Exercise 2.8

Which will deliver a higher future value after one year, a deposit of $\$1,000$ attracting interest at 15% compounded daily, or at 15.5% compounded semi-annually?

Exercise 2.9

What initial investment subject to annual compounding at 12% is needed to produce $\$1,000$ after two years?

The last exercise touches upon the problem of finding the present value of an amount payable at some future time instant in the case when periodic compounding applies. Here the formula for the *present* or *discounted value* of $V(t)$ is

$$V(0) = V(t)(1+\frac{r}{m})^{-tm},$$

the number $(1+\frac{r}{m})^{-tm}$ being the *discount factor*.

Remark 2.1

Fix the terminal value $V(t)$ of an investment. It is an immediate consequence of Proposition 2.1 that the present value increases if any one of the factors r, t, m decreases, the other ones remaining unchanged.

Exercise 2.10

Find the present value of \$100,000 to be received after 100 years if the interest rate is assumed to be 5% throughout the whole period and a) daily or b) annual compounding applies.

One often requires the value $V(t)$ of an investment at an intermediate time $0 < t < T$, given the value $V(T)$ at some fixed future time T. This can be achieved by computing the present value of $V(T)$, taking it as the principal, and running the investment forward up to time t. Under periodic compounding with frequency m and interest rate r, this obviously gives

$$V(t) = \left(1 + \frac{r}{m}\right)^{-(T-t)m} V(T). \tag{2.6}$$

To find the return on a deposit attracting interest compounded periodically we use the general formula (2.3) and readily arrive at

$$K(s,t) = \frac{V(t) - V(s)}{V(s)} = (1 + \frac{r}{m})^{(t-s)m} - 1.$$

In particular,

$$K(0, \frac{1}{m}) = \frac{r}{m},$$

which provides a simple way of computing the interest rate given the return.

Exercise 2.11

Find the return over one year under monthly compounding with $r = 10\%$.

Exercise 2.12

Which is greater, the interest rate r or the return $K(0,1)$ if the compounding frequency m is greater than 1?

Remark 2.2

The return on a deposit subject to periodic compounding is *not* additive. Take, for simplicity, $m = 1$. Then

$$K(0,1) = K(1,2) = r,$$
$$K(0,2) = (1+r)^2 - 1 = 2r + r^2,$$

and clearly $K(0,1) + K(1,2) \neq K(0,2)$.

2.1.3 Streams of Payments

An *annuity* is a sequence of finitely many payments of a fixed amount due at equal time intervals. Suppose that payments of an amount C are to be made once a year for n years, the first one due a year hence. Assuming that annual compounding applies, we shall find the present value of such a stream of payments. We compute the present values of all payments and add them up to get

$$\frac{C}{1+r} + \frac{C}{(1+r)^2} + \frac{C}{(1+r)^3} + \cdots + \frac{C}{(1+r)^n}.$$

It is sometimes convenient to introduce the following seemingly cumbersome piece of notation:

$$\mathrm{PA}(r,n) = \frac{1}{1+r} + \frac{1}{(1+r)^2} + \cdots + \frac{1}{(1+r)^n}. \tag{2.7}$$

This number is called the *present value factor for an annuity*. It allows us to express the present value of an annuity in a concise form:

$$\mathrm{PA}(r,n) \times C.$$

The expression for $\mathrm{PA}(r,n)$ can be simplified by using the formula

$$a + qa + q^2a + \cdots + q^{n-1}a = a\frac{1-q^n}{1-q}.$$

In our case $a = \frac{1}{1+r}$ and $q = \frac{1}{1+r}$, hence

$$\mathrm{PA}(r,n) = \frac{1-(1+r)^{-n}}{r}. \tag{2.8}$$

Remark 2.3

Note that an initial bank deposit of

$$P = \mathrm{PA}(r,n) \times C = \frac{C}{1+r} + \frac{C}{(1+r)^2} + \cdots + \frac{C}{(1+r)^n}$$

attracting interest at a rate r compounded annually would produce a stream of n annual payments of C each. A deposit of $C(1+r)^{-1}$ would grow to C after one year, which is just what is needed to cover the first annuity payment. A deposit of $C(1+r)^{-2}$ would become C after two years to cover the second payment, and so on. Finally, a deposit of $C(1+r)^{-n}$ would deliver the last payment of C due after n years.

Example 2.4

Consider a loan of $\$1,000$ to be paid back in 5 equal instalments due at yearly intervals. The instalments include both the interest payable each year calculated at 15% of the current outstanding balance and the repayment of a fraction of the loan. A loan of this type is called an *amortised loan*. The amount of each instalment can be computed as

$$\frac{1,000}{\mathrm{PA}(15\%, 5)} \cong 298.32.$$

This is because the loan is equivalent to an annuity from the point of view of the lender.

Exercise 2.13

What is the amount of interest included in each instalment? How much of the loan is repaid as part of each instalment? What is the outstanding balance of the loan after each instalment is paid?

Exercise 2.14

How much can you borrow if the interest rate is 18%, you can afford to pay $\$10,000$ at the end of each year, and you want to clear the loan in 10 years?

Exercise 2.15

Suppose that you deposit $\$1,200$ at the end of each year for 40 years, subject to annual compounding at a constant rate of 5%. Find the balance after 40 years.

Exercise 2.16

Suppose that you took a mortgage of $\$100,000$ on a house to be paid back in full by 10 equal annual instalments, each consisting of the interest due on the outstanding balance plus a repayment of a part of the amount borrowed. If you decided to clear the mortgage after eight years, how much money would you need to pay on top of the 8th instalment, assuming that a constant annual compounding rate of 6% applies throughout the period of the mortgage?

Recall that a *perpetuity* is an infinite sequence of payments of a fixed amount C occurring at the end of each year. The formula for the present value of a

perpetuity can be obtained from (2.7) in the limit as $n \to \infty$:

$$\lim_{n\to\infty} \mathrm{PA}(r,n) \times C = \frac{C}{1+r} + \frac{C}{(1+r)^2} + \frac{C}{(1+r)^3} + \cdots = \frac{C}{r}. \tag{2.9}$$

The limit amounts to taking the sum of a geometric series.

Remark 2.4

The present value of a perpetuity is given by the same formula as in Example 2.2, even though periodic compounding has been used in place of simple interest. In both cases the annual payment C is exactly equal to the interest earned throughout the year, and the amount remaining to earn interest in the following year is always $\frac{C}{r}$. Nevertheless, periodic compounding allows us to view the same sequence of payments in a different way: The present value $\frac{C}{r}$ of the perpetuity is decomposed into infinitely many parts, as in (2.9), each responsible for producing one future payment of C.

Remark 2.5

Formula (2.8) for the annuity factor is easier to memorise in the following way, using the formula for a perpetuity: The sequence of n payments of $C = 1$ can be represented as the difference between two perpetuities, one starting now and the other after n years. (Cutting off the tail of a perpetuity, we obtain an annuity.) In doing so we need to compute the present value of the latter perpetuity. This can be achieved by means of the discount factor $(1+r)^{-n}$. Hence,

$$\mathrm{PA}(r,n) = \frac{1}{r} - \frac{1}{r} \times \frac{1}{(1+r)^n} = \frac{1-(1+r)^{-n}}{r}.$$

Exercise 2.17

Find a formula for the present value of an infinite stream of payments of the form C, $C(1+g)$, $C(1+g)^2, \ldots$, growing at a constant rate g. By the tail-cutting procedure find a formula for the present value of n such payments.

2.1.4 Continuous Compounding

Formula (2.5) for the future value at time t of a principal P attracting interest at a rate $r > 0$ compounded m times a year can be written as

$$V(t) = \left[\left(1 + \frac{r}{m}\right)^{\frac{m}{r}}\right]^{tr} P.$$

In the limit as $m \to \infty$, we obtain

$$V(t) = \mathrm{e}^{tr} P, \tag{2.10}$$

where

$$\mathrm{e} = \lim_{x \to \infty} \left(1 + \frac{1}{x}\right)^{x}$$

is the base of natural logarithms. This is known as *continuous compounding*. The corresponding *growth factor* is e^{tr}. A typical graph of $V(t)$ is shown in Figure 2.3.

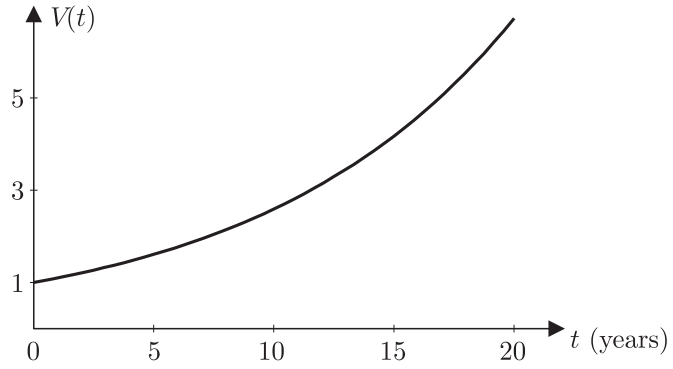


Figure 2.3 Continuous compounding at 10% ($r = 0.1$, $P = 1$)

The derivative of $V(t) = \mathrm{e}^{tr} P$ is

$$V'(t) = r\mathrm{e}^{tr} P = rV(t).$$

In the case of continuous compounding the rate of the growth is proportional to the current wealth.

Formula (2.10) is a good approximation of the case of periodic compounding when the frequency m is large. It is simpler and lends itself more readily to transformations than the formula for periodic compounding.

Exercise 2.18

How long will it take to earn \$1 in interest if \$1,000,000 is deposited at 10% compounded continuously?

Exercise 2.19

In 1626 Peter Minuit, governor of the colony of New Netherland, bought the island of Manhattan from Indians paying with beads, cloth, and trinkets worth \$24. Find the value of this sum in year 2000 at 5% compounded a) continuously and b) annually.

Proposition 2.2

Continuous compounding produces higher future value than periodic compounding with any frequency m, given the same initial principal P and interest rate r.

Proof

It suffices to verify that

$$\mathrm{e}^{tr} > (1+\frac{r}{m})^{tm} = \left[(1+\frac{r}{m})^{\frac{m}{r}}\right]^{rt}.$$

The inequality holds because the sequence $(1+\frac{r}{m})^{\frac{m}{r}}$ is increasing and converges to e as $m \nearrow \infty$. □

Exercise 2.20

What will be the difference between the value after one year of \$100 deposited at 10% compounded monthly and compounded continuously? How frequent should the periodic compounding be for the difference to be less than \$0.01?

The present value under continuous compounding is obviously given by

$$V(0) = V(t)\mathrm{e}^{-tr}.$$

In this case the *discount factor* is e^{-tr}. Given the terminal value $V(T)$, we clearly have

$$V(t) = \mathrm{e}^{-r(T-t)}V(T). \tag{2.11}$$

Exercise 2.21

Find the present value of $\$1,000,000$ to be received after 20 years assuming continuous compounding at 6%.

Exercise 2.22

Given that the future value of \$950 subject to continuous compounding will be $\$1,000$ after half a year, find the interest rate.

The return $K(s,t)$ defined by (2.3) on an investment subject to continuous compounding fails to be additive, just like in the case of periodic compounding. It proves convenient to introduce the *logarithmic return*

$$k(s,t) = \ln \frac{V(t)}{V(s)}. \tag{2.12}$$

Proposition 2.3

The logarithmic return is additive,

$$k(s,t) + k(t,u) = k(s,u).$$

Proof

This is an easy consequence of (2.12):

$$\begin{aligned} k(s,t) + k(t,u) &= \ln \frac{V(t)}{V(s)} + \ln \frac{V(u)}{V(t)} \\ &= \ln \frac{V(t)}{V(s)} \frac{V(u)}{V(t)} = \ln \frac{V(u)}{V(s)} = k(s,u). \end{aligned}$$

□

If $V(t)$ is given by (2.10), then $k(s,t) = r(t-s)$, which enables us to recover the interest rate

$$r = \frac{k(s,t)}{t-s}.$$

Exercise 2.23

Suppose that the logarithmic return over 2 months on an investment subject to continuous compounding is 3%. Find the interest rate.

2.1.5 How to Compare Compounding Methods

As we have already noticed, frequent compounding will produce a higher future value than less frequent compounding if the interest rates and the initial principal are the same. We shall consider the general circumstances in which one compounding method will produce either the same or higher future value than another method, given the same initial principal.

Example 2.5

Suppose that certificates promising to pay \$120 after one year can be purchased or sold now, or at any time during this year, for \$100. This is consistent with a constant interest rate of 20% under annual compounding. If an investor decided to sell such a certificate half a year after the purchase, what price would it fetch? Suppose it is \$110, a frequent first guess based on halving the annual profit of \$20. However, this turns out to be too high a price, leading to the following arbitrage strategy:

- Borrow $\$1,000$ to buy 10 certificates for \$100 each.
- After six months sell the 10 certificates for \$110 each and buy 11 new certificates for \$100 each. The balance of these transactions is nil.
- After another six months sell the 11 certificates for \$110 each, cashing $\$1,210$ in total, and pay $\$1,200$ to clear the loan with interest. The balance of \$10 would be the arbitrage profit.

A similar argument shows that the certificate price after six months cannot be too low, say, \$109.

The price of a certificate after six months is related to the interest rate under semi-annual compounding: If this rate is r, then the price is $100\left(1+\frac{r}{2}\right)$ dollars and vice versa. Arbitrage will disappear if the corresponding growth factor $\left(1+\frac{r}{2}\right)^2$ over one year is equal to the growth factor 1.2 under annual compounding,

$$\left(1+\frac{r}{2}\right)^2 = 1.2,$$

which gives $r \cong 0.1909$, or 19.09%. If so, then the certificate price after six months should be $100\left(1+\frac{0.1909}{2}\right) \cong 109.54$ dollars.

The idea based on considering the growth factors over a fixed period, typically one year, can be used to compare any two compounding methods.

Definition 2.1

We say that two compounding methods are *equivalent* if the corresponding growth factors over a period of one year are the same. If one of the growth factors exceeds the other, then the corresponding compounding method is said to be *preferable*.

Example 2.6

Semi-annual compounding at 10% is equivalent to annual compounding at 10.25%. Indeed, in the former case the growth factor over a period of one year is

$$\left(1+\frac{0.1}{2}\right)^2 = 1.1025,$$

which is the same as the growth factor in the latter case. Both are preferable to monthly compounding at 9%, for which the growth factor over one year is only

$$\left(1+\frac{0.09}{12}\right)^{12} \cong 1.0938.$$

We can freely switch from one compounding method to another equivalent method by recalculating the interest rate. In the chapters to follow we shall normally use either annual or continuous compounding.

Exercise 2.24

Find the rate for continuous compounding equivalent to monthly compounding at 12%.

Exercise 2.25

Find the frequency of periodic compounding at 20% to be equivalent to annual compounding at 21%.

Instead of comparing the growth factors, it is often convenient to compare the so-called effective rates as defined below.

Definition 2.2

For a given compounding method with interest rate r the *effective rate* r_{e} is one that gives the same growth factor over a one year period under annual compounding.

In particular, in the case of periodic compounding with frequency m and rate r the effective rate r_{e} satisfies

$$\left(1+\frac{r}{m}\right)^m = 1 + r_{\mathrm{e}}.$$

In the case of continuous compounding with rate r

$$\mathrm{e}^r = 1 + r_{\mathrm{e}}.$$

Example 2.7

In the case of semi-annual compounding at 10% the effective rate is 10.25%, see Example 2.6.

Proposition 2.4

Two compounding methods are equivalent if and only if the corresponding effective rates r_{e} and r'_{e} are equal, $r_{\mathrm{e}} = r'_{\mathrm{e}}$. The compounding method with effective rate r_{e} is preferable to the other method if and only if $r_{\mathrm{e}} > r'_{\mathrm{e}}$.

Proof

This is because the growth factors over one year are $1 + r_{\mathrm{e}}$ and $1 + r'_{\mathrm{e}}$, respectively. □

Example 2.8

In Exercise 2.8 we have seen that daily compounding at 15% is preferable to semi-annual compounding at 15.5%. The corresponding effective rates r_{e} and r'_{e} can be found from

$$1 + r_{\mathrm{e}} = \left(1+\frac{0.15}{365}\right)^{365} \cong 1.1618,$$
$$1 + r'_{\mathrm{e}} = \left(1+\frac{0.155}{2}\right)^{2} \cong 1.1610.$$

This means that r_{e} is about 16.18% and r'_{e} about 16.10%.

Remark 2.6

Recall that formula (2.5) for periodic compounding, that is,

$$V(t) = \left(1+\frac{r}{m}\right)^{tm} P,$$

admits only time instants t being whole multiples of the compounding period $\frac{1}{m}$. An argument similar to that in Example 2.5 shows that the appropriate no-arbitrage value of an initial sum P at any time $t \geq 0$ should be $\left(1 + \frac{r}{m}\right)^{tm} P$. A reasonable extension of (2.5) is therefore to use the right-hand side for all $t \geq 0$ rather than just for whole multiples of $\frac{1}{m}$. From now on we shall always use this extension.

In terms of the effective rate r_{e} the future value can be written as

$$V(t) = (1 + r_{\mathrm{e}})^t P.$$

for all $t \geq 0$. This applies both to continuous compounding and to periodic compounding extended to arbitrary times as in Remark 2.6. Proposition 2.4 implies that, given the same initial principal, equivalent compounding methods will produce the same future value for all times $t \geq 0$. Similarly, a compounding method preferable to another one will produce a higher future value for all $t > 0$.

Remark 2.7

Simple interest does not fit into the scheme for comparing compounding methods. In this case the future value $V(t)$ is a linear function of time t, whereas it is an exponential function if either continuous or periodic compounding applies. The graphs of such functions have at most two intersection points, so they can never be equal to one another for all times $t \geq 0$ (except for the trivial case of zero principal).

Exercise 2.26

What is the present value of an annuity consisting of monthly payments of an amount C continuing for n years? Express the answer in terms of the effective rate r_{e}.

Exercise 2.27

What is the present value of a perpetuity consisting of bimonthly payments of an amount C? Express the answer in terms of the effective rate r_{e}.